Assignment 1

Problem Set 2

Anindya Biswas

Roll No : 002420701050, UG-2, Physics

if-else conditions:

1. Calculate BMI and print whether underweight, normal weight, overweight, obese

In [10]:
```
a=input("Type your weight in kg: ")
b=input("Type your height in metre: ")
weight=float(a)
height=float(b)
bmi=weight/(height*height)
print("Your BMI is: ", bmi)
if bmi < 18.5:
    print("Your category is: Underweight")
if bmi > 18.5 and bmi <=24.9:
    print("Your category is: Normal weight")
if bmi > 24.9 and bmi <=29.9 :
    print("Your category is: Overweight")
if bmi > 29.9 :
    print("Your category is: Obese")
```

```
Your BMI is:  17.68978885166924
Your category is: Underweight
```

2. Paper, rock, scissor game with computer

In [40]:
```
import random
def game():
    options=["rock", "paper", "scissor"]
    print ("Your options are: ", options)
    user_choice=input('Type your choice: ')
    if user_choice not in options:
        print('Invalid input! please try again!')
        return
    comp_choice=random.choice(options)
    print('The Computer chose: ', comp_choice)
    if user_choice==comp_choice :
        print("This is a tie!")
    elif (user_choice=="rock" and comp_choice=="scissor") or \
         (user_choice=="paper" and comp_choice=="rock") or \
         (user_choice=="scissor" and comp_choice=="paper") :
        print ('You won!')
    else:
        print("Computer won! Better luck next time.")
game()
```

```
Your options are:  ['rock', 'paper', 'scissor']
The Computer chose:  paper
Computer won! Better luck next time.
```

3. Write a Python program that asks the user to input a number and then checks if the number is even or odd.

In [102…

```
def odd_even():
    a=input("Type any integer: ")
    if '.' in a:
        print("The number is not an integer, please provide an integer va
        return
    a=int(a)
    if a%2==0:
        print("The number is even!")
    else:
        print("the number is odd!")
odd_even()
```

```
the number is odd!
```

4. Write a Python program that checks if a given year is a leap year.

In [86]:

```
year=int(input("Type any year: "))
if (year%400==0) or (year%100!=0 and year%4==0):
    print("The year is a leap year")
else :
    print("The year is a not leap year")
```

```
The year is a leap year
```

5. Write a program that asks the user to input two numbers and a mathematical operation (+, -, *, /). The program then performs the operation and outputs the result. If the user enters an invalid operation, print an error message.

In [89]:

```
a=float(input("Type the 1st number: "))
b=float(input("Type the 2nd number: "))
options=['+','-','*','/']
print("Your operations are: ", options)
c=input("Choose your operation: ")
if c=='+':
    print("The sum is: ", float(a+b))
elif c=='-':
    print("The subtraction is: ", float(a-b))
elif c=='*':
    print("The product is: ", float(a*b))
elif c=='/':
    print("The divison is: ", float(a/b))
```

```
Your operations are:  ['+', '-', '*', '/']
The divison is:  0.5
```

“for-while” loop

6. Take a number as input and print its factorial using both for and while loops.

Using while loop

In [5]:
```
n=int(input("Type any integer to find its factorial: "))

def func(n):
    fact=1
    i=1
    while i<=n:
        fact*=i
        i+=1
    return fact

print("The value of the factorial is: ",func(n))
```

```
The value of the factorial is:  120
```

Using for loop

In [18]:
```
n=int(input("Type any integer to find its factorial: "))
def f(n):
    fact=1
    i=1
    for i in range(1,n+1,1):
        fact=fact*i
    return fact
print("The value of the factrial is: ", f(n))
```

```
The value of the factrial is:  5040
```

7. Write a Python program that checks if a number n is prime or not. Use "for" loop.

In [36]:
```
n=int(input("Type any integer: "))

def f(n):
    if n<2:
        print("The number is not a Prime Number!")
        return
    for i in range(2,n,1):
        if n%i==0:
            print("The number is not prime!")
            return
    print("The number is prime!")
f(n)
```

```
The number is prime!
```

8. Write a program that prints the first n numbers of the Fibonacci sequence using a for loop.

In [44]:
```
n=int(input("Type any whole number: "))
def f():
    i=0
    sum=0
    print("The fibonacci series will be: ")
    for i in range(0,n,1):
        sum=sum+i
        print(sum)
```

```
    return
f()
```

```
The fibonacci series will be:
0
1
3
6
10
15
```

9. Bisection Method for Root Finding: Find the root of a function using the bisection method. Use a while loop to iterate until the root is found within a given tolerance.

In [1]:
```
#initial guess
a=float(input('Type any guess :'))
b=float(input('Type another guess:'))
tol=0.000000000000001
def f(x):
    return x*x-4

if f(a)*f(b)>0:
   print("The number is not in your interval")
else:
    while abs(b-a)>tol:
        c=(a+b)/2
        if f(c) == 0:
            print("The root is :", c)
        elif f(a) * f(c) <= 0:
            b = c
        else:
            a = c
print(c)
```

```
1.9999999999999991
```

10. Newton’s Method for Solving Equations: Implement Newton-Raphson method to find roots of a given function. Use a while loop to update the estimate until convergence.

In [8]:
```
x=float(input("Type any guess"))
def f(x):
    return x*x-4 # my polynomial here
def df(x):
    return 2*x #derivative of the polynomial
while abs(f(x))>0.000000000001:
    x=x-f(x)/df(x)
print("The root is : ", x )
```

```
The root is :  2.0
```

11. Take a number as input and print its multiplication table up to 10.

In [49]:
```
n=int(input("Type any integer: "))
print("The multipliation table will be: ")
for i in range(1,11,1):
    a=n*i
    print(a)
```

```
The multipliation table will be:
5
10
15
20
25
30
35
40
45
50
```